10 — Counterstrategy

Mununu Tutorial / Mariano Cerrutti

Formula Inversion

Duality table — swap operators to invert a formula:

Controller	⟷	Environment
`mu`	↔	`nu`
`nu`	↔	`mu`
`[]` (box)	↔	`<>` (diamond)
`<>` (diamond)	↔	`[]` (box)
`&&`	↔	`||`
`||`	↔	`&&`
`ctrl = controllable`	↔	`ctrl = environment`

Inverting a formula swaps the roles of controller and environment: the environment becomes the "player" trying to satisfy the inverted property.

Syntax

A counterstrategy formula targets the *complement* of the controller's goal:

```
body = nu X. ((!target) &&
    < (ctrl = environment) > X);
```

This computes the set of states from which the environment can **avoid** `target` forever.

Environment Diamond

`< (ctrl = environment) > phi` means:

- There **exists** an uncontrollable transition leading to φ, OR
- **All** controllable transitions lead to φ.

The environment "wins" a step if it can force φ via an uncontrollable move, or if every controllable option the controller might pick also leads to φ.

Skolem Paradigm

Nondeterminism is adversarial for **both** players. The controller chooses a label, not a transition; similarly, the environment chooses a label. Multiple targets under the same label are resolved adversarially against the choosing player.

Common Patterns

Env prevents target:

```
nu X. ((!target) &&
    < (ctrl = environment) > X)
```

Env counterstrategy (liveness):

```
mu X. (bad || < (ctrl = environment) > X)
```

Environment can force the system into `bad`.

CLI

Generate a counterstrategy graph:

```
mununu context graph file.ctxdsl \
    --counterstrategy \
    --formula F \
    --automaton A \
    --output cs.html
```

The output highlights the environment's winning region and the transitions it uses to defeat the controller.

Web UI

- **Verification tab** — click the *Counterstrategy* button to visualize the environment winning region.
- **Synthesis tab** — full diagnostics including counterexample traces and counterstrategy graphs.

References

Zielonka (1998) "Infinite Games on Finitely Coloured Graphs with Applications to Automata on Infinite Trees" — parity game solving. Bruse, Friedmann & Lange (2016) — strategy extraction from fixpoint iteration. Emerson & Jutla (1991) — positional determinacy of parity games.